Brain MRI Tumor Segmentation and Quantitative Analysis Using Classical Image Processing

Week 8 - Removing Border False Detections and Improving Tumor Volume Estimation

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Submitted By:- G9 - The Medics.

I. Work Done This Week:-

This week we worked on fixing the two main problems we had planned to solve at the end of Week 7. In last week's report, moving from raw subtraction to a statistical Z-score baseline atlas and Difference of Gaussians (DoG) improved our results, but we still had unwanted garbage values appearing on the outer border of the brain and inside the dark fluid spaces (ventricles), and for several scans the detected tumor volume did not match the actual tumor size.

When we analyzed why the border garbage values were happening, we found that different patient scans include different amounts of the neck and lower head, which shifts the actual brain up or down by several slices even when the 3D matrix dimensions are the same. In addition, morphological skull stripping was occasionally cutting off dark tumors located right on the outer edge of the brain. The updated mathematical approach is defined below.

A. 1. Bilateral Brain Masking and Brain-Box Alignment

During skull stripping with 3D erosion and dilation, a dark necrotic tumor touching the outer surface of the cortex has low intensity similar to cerebrospinal fluid and can get accidentally removed along with the skull. Since a brain tumor grows on one hemisphere while the opposite hemisphere still has a normal outer boundary, we mirror our initial brain mask $M(x, y, z)$ across the mid-sagittal plane ($x = D_x/2$) and take the union of both sides:

$$M_{\text{full}}(x, y, z) = M(x, y, z) \cup M(D_x - x,\ y,\ z). \tag{1}$$

Next, instead of aligning the entire image box (which includes varying neck slices), we find the tight 3D bounding box $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}]$ of the isolated brain mask M_{full} and map every brain into the exact same standard brain coordinate box using trilinear interpolation:

$$x' = a_x x + b_x, \quad y' = a_y y + b_y, \quad z' = a_z z + b_z. \tag{2}$$

This ensures the top of the cortex and bottom of the brain line up properly with the baseline atlas and eliminates the border garbage regions from Week 7.

B. 2. Combining Baseline and Bilateral Asymmetry Scoring

Using our 7 tumor-free baseline scans and their left-right reflections ($M = 14$ aligned healthy brain volumes), we compute the normal mean intensity $\bar{B}(x, y, z)$ and standard deviation $\sigma_B(x, y, z)$ at every voxel:

$$\bar{B}(x, y, z) = \frac{1}{M} \sum_{k=1}^{M} B_k(x, y, z). \tag{3}$$

We suppress natural dark fluid regions inside the central ventricles near the midline. Then, for every brain voxel $I(x, y, z)$, we measure its normalized darkness compared to the healthy baseline ($\bar{B}$) and compared to the opposite side of the same patient's brain (I_{flip}):

$$D_{\text{base}} = \frac{\max(0,\ \bar{B} - I)}{\sigma_B + \epsilon}, \qquad D_{\text{side}} = \frac{\max(0,\ I_{\text{flip}} - I)}{\sigma_{\text{side}} + \epsilon}. \tag{4}$$

Because a true tumor is abnormal relative to both the healthy population baseline and the patient's own opposite hemisphere, we combine both into a joint anomaly score:

$$S(x, y, z) = \sqrt{D_{\text{base}}(x, y, z) \cdot D_{\text{side}}(x, y, z)}. \tag{5}$$

Comparing the total score between the left and right hemispheres locks onto the tumor-bearing side and locates the 3D tumor seed point.

C. 3. Adaptive Region Growing and Volumetric Estimation

Using one fixed threshold to grow the tumor outline failed when switching between sharp, dark cavities and faint, spread-out tumors. To adapt to each scan, we compare the mean intensity inside the candidate tumor core (μ_{tumor}) against the opposite healthy hemisphere (μ_{healthy}) along with the local intensity spread σ_I:

$$\Delta I = \mu_{\text{healthy}} - \mu_{\text{tumor}}, \qquad V = N_v \times C_t. \tag{6}$$

When ΔI and σ_I are large, a stricter threshold prevents the region from leaking into adjacent sulci; when the tumor is faint, a relaxed threshold captures the full infiltrative margin before 3D Gaussian surface smoothing and physical volume calculation (V) from voxel count C_t and voxel volume N_v.

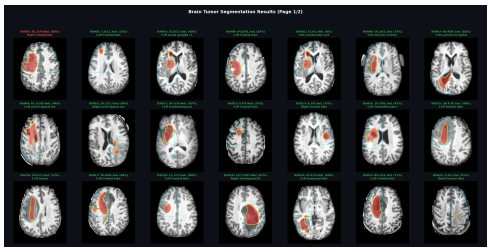


Fig. 1. Week 8 Results Pg1

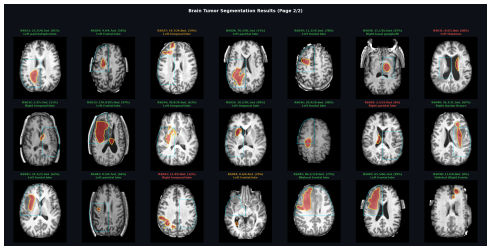


Fig. 2. Week 8 Results Pg2

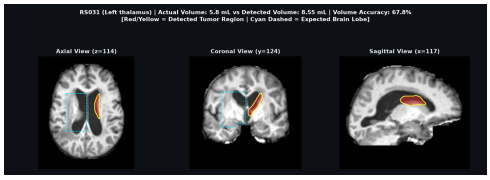


Fig. 3. False Result

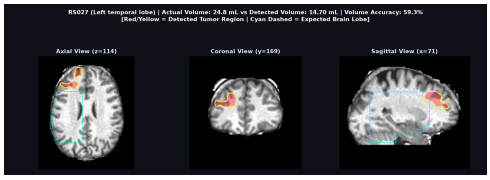


Fig. 4. Partially True Result

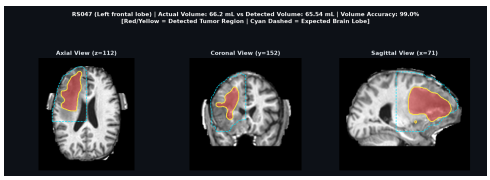


Fig. 5. True Result

TABLE I
SUMMARY OF DETECTION AND VOLUME ACCURACY ACROSS THE DATASET.

Evaluation Metric	Unique Scans (39)	All Scans (41)
Correct Hemisphere Detected	**35 / 39 (89.7%)**	37 / 41 (90.2%)
Correct Brain Lobe Detected	**33 / 39 (84.6%)**	35 / 41 (85.4%)
Average Volume Accuracy	**70.7%**	68.3%
Combined Location & Volume Accuracy	**78.9%**	78.1%

II. WORK TO BE DONE NEXT WEEK:-

As shown in the figures and summary table above, aligning the isolated brain bounding box and adapting the region-growing threshold using the opposite healthy hemisphere removed the outer border garbage values from Week 7 and improved our accuracy across the dataset. However, while many scans now produce accurate locations and volumes (such as Fig. 5), there are still some false reports of tumor detection (such as Fig. 3, where a tumor near the midline pushes the central fluid cavities) and partial volume errors on low-contrast tumors (such as Fig. 4) that still need to be fixed and improved.

We are now focusing on improving our approach to eliminate these remaining false detections and volume mismatches so we can produce the best results. As we have started working on the GUI as well, we are planning to complete it and integrate it with our system for a seamless user experience. We are also finding another dataset to use as a reference point, and our first priority is to use a dataset that has binary ground-truth masking on the segmented tumor areas so that we can compare our results directly against it for a clear voxel-by-voxel comparison. This will help us derive the exact precision of our approach. Next week will be the last week of our project, so we are checking on every aspect to make sure that we have not missed any point.

Repo Link:-
https://github.com/Rushin-Mod/DIP_2026_P20_G9↗